Quantitative constraints on the scope of negative selection

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Maturing T cells with a high affinity for self-antigens presented in the thymus are deleted in the process of negative selection. Although the expression of various 'tissue-specific' antigens has been described in the thymus, it is still controversial what fraction of all self-antigens induces tolerance by this mechanism. We demonstrate that the limited duration of the negative selection phase imposes a constraint on the number of self-peptides that can be reliably selected against. The analysis supports the theory that negative selection is confined to the subset of peptides produced by dendritic cells.

The classical mechanism for central tolerance is the deletion of autoreactive T cells in the thymus by the process of negative selection [1]. T cells showing too high an affinity for self-peptide–MHC complexes (spMHCs) displayed on the surface of antigen-presenting cells (APCs) undergo apoptosis during this crucial period. Theoretical considerations have suggested that only T cells specific for spMHCs normally present on professional APCs in the periphery should be deleted [2]. It is, however, not clear what fraction of all spMHCs find their way to the surface of peripheral APCs [3,4], let alone into the thymus. Until recently, the thymus had been thought to be accessible only to ubiquitously expressed proteins and abundant blood-borne self-antigens. Accumulating evidence, however, indicates that a promiscuous expression of 'tissue-specific' antigens constitutes an 'immunological self-shadow' in the thymus [5–8]. In this Opinion we investigate whether all self-antigens could, in principle, be reliably selected against by such a mechanism.

Negative selection cannot involve all self-peptides

The efficiency of negative selection can be characterized by the probability that an autoreactive thymocyte escapes negative selection and leaves the thymus as a naïve T cell. During the process of negative selection, maturing thymocytes scan the surface of APCs, mostly dendritic cells (DCs), in the thymus [1]. We assess whether DCs, the most potent mediators of negative selection [9–11], could select against all spMHCs reliably, if they present all spMHCs with an equal probability. (Note that the assumption of equal presentation is conservative because any deviation from this assumption will result in a lower total number of distinct spMHC presented per cell.) Therefore, if DCs cannot select reliably against all spMHCs with this relaxed assumption, the conclusion is robust, and neither can other cell types, such as medullary thymic epithelial cells (mTECs) with a smaller presentation capacity, accomplish the task.

If an autoreactive cell encounters its cognate spMHC, it is induced into apoptosis and dies [1]. That is, the cell survives only if the cognate spMHC is not presented to it by any of the APCs that it scans during negative selection. If each maturing cell scans n APCs, and P is the probability that any one APC presents the cognate spMHC, the probability of escaping negative selection can be written as $P_E = (1 - P)^n$. The average probability that a self-peptide is presented on an APC can be written as $P = C_P/C_T$, where C_P is the number of spMHC presented by one APC for negative selection and C_T is the total number of distinct spMHCs. The threshold spMHC density required to mediate negative selection has been estimated as 1500 and 16 000 complexes for two known pMHC associations [11]. However, naïve T cells might respond to much lower surface densities of their agonist ligand at least in some experimental settings [12]. Furthermore, the threshold for negative selection in the thymus might be smaller than the threshold for activation in the periphery [13] and incomplete tolerance could be achieved with the presentation of ≤ 100 complexes per APC [14]. We have therefore considered thresholds of 1500, 150 and 15 complexes. With a total of 5×10^5 MHC molecules on their surface [11], DCs can present for selection at most $Cp = 5 \times 10^5/1500 \cong 333$ distinct complexes with a selection threshold of 1500 complexes. For thresholds of 150 and 15 molecules, 10-fold and 100-fold more complexes could be presented, respectively. Note that these estimates are strict upper bounds in that they assume that complexes with the given threshold occupy the whole presentation capacity of the DC.

Next, we calculate the total number of distinct MHC class-I restricted spMHCs. The complications of MHC class-II restricted presentation are discussed at the end of this Opinion. Most binding peptides contain 8–10 amino acids [15], and yield $\sim 3 \times 10^7$ peptides of presentable length in the mouse proteome (downloaded from the SWISS-PROT/TrEMBL database at http://www.ebi.ac.uk/proteome/MOUSE/download.html). However, not all self-peptides are able to form spMHCs. Proteasomes generate $\sim 30\%$ of all possible peptides [16] and only a fraction of all produced peptides might bind to MHC molecules. Prediction based on known epitopes for the mouse H-2 I-E^k MHC molecule has estimated that 3% of all possible nonameric peptides can be presented in the context of this molecule

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[17]. To obtain the ratio of presented complexes to all peptides of presentable length, we have to sum up the probabilities of presentation on each of the alleles. In mice, a heterozygous individual expresses six MHC class-I alleles [18], which increases the fraction of total peptides that are presented to $0.03 \times 6 = 0.18$. Thus, we estimate that there are $C_T = 0.18 \times 3 \times 10^7 = 5.4 \times 10^6$ distinct spMHCs. A selection threshold of 1500 molecules gives $P = C_P/C_T = 333/5.4 \times 10^6 \cong 6 \times 10^{-5}$ for the average probability that a given spMHC is presented by a given DC in the thymus. With thresholds of 150 and 15 molecules, the probability is 6×10^{-4} and 6×10^{-3}, respectively.

The number of DCs that a thymocyte scans, n, can be estimated as the total length of the negative selection phase divided by the time needed to scan one DC. The thymic maturation of thymocytes takes about three weeks [18]. Negative selection probably takes place in the thymic medulla, where thymocytes spend ~5–14 days [19]. Note that selection needs to act efficiently even during the shortest time in the interval to ensure the purging of the repertoire. Furthermore, the >100-fold excess of thymocytes over DCs in the thymus [9,19] probably restricts the effective time available for scanning because not all thymocytes can be served simultaneously. 14 days (2×10^4 mins) can therefore be regarded as an upper limit for the time available. The time needed for a thymocyte to scan one DC has been estimated to be 6–12 mins in experimental systems mirroring *in vivo* DC architecture [20], consistent with *in vivo* observations on the random walk of T cells in extracted lymph nodes [21]. Assuming a 5 min. scanning time, we derive an upper bound for the number of DCs scanned in 14 days as $n = 2 \times 10^4/5 = 4000$. With this approximation, an autoreactive thymocyte can escape selection with a probability $P_E = (1-P)^n = (1 - 6 \times 10^{-5})^{4000} \cong 0.79$, for a selection threshold of 1500 molecules. For thresholds of 150 and 15 complexes, we obtain probabilities of 0.09 and 3.8×10^{-11}, respectively. The calculations and parameter estimates are summarized in Table 1.

Our calculation predicts a surprisingly high level of autoimmunity: depending on the threshold for selection, ≤79% of autoreactive thymocytes might not be confronted with their cognate spMHC during selection and might hence leave the thymus unopposed. Even with a threshold of 150 molecules, 9% of all autoreactive cells could escape deletion. Negative selection could be effective on all spMHCs only if the vast majority of the complexes were never presented in more than a few dozen copies on peripheral DCs. Indeed, autoreactive T cells are present in the periphery [22,23], yet autoimmune diseases are rare. How can we resolve this paradox? Either all self-peptides are involved in negative selection, in which case the residual autoreactivity must be removed or neutralized by another mechanism that covers the whole array. Alternatively, negative selection acts on a subset of all self-peptides, on which it can be very efficient. In this case, the auxiliary mechanisms of peripheral tolerance [24] would cover autoreactivity against the self-peptides not included in the subset. We now argue for a restricted scope of negative selection based on quantitative considerations.

Size and nature of the selecting repertoire

The number of spMHCs mediating negative selection can be deduced from the overall probability of thymocytes surviving negative selection (Box 1). By substituting the estimate of 2.2×10^5 selecting spMHCs for C_T, we find that a thymocyte reactive to an spMHC within this subset escapes negative selection with a probability as low as 3×10^{-27} (for an intermediate selection threshold of 150 complexes). The high reliability implies that virtually all complexes involved in negative selection will be presented to all thymocytes. Negative selection is thus restricted to, and is efficient on, a small subset of all spMHCs. Matzinger proposed almost ten years ago that tolerance by deletion should be induced by the peptides displayed on professional APCs [2]. Mature DCs are the dedicated cell type of the immune system, able to induce the activation of naïve cells [25]. Self-peptides produced by DCs are presented in an activating context at all sites of infection or in the presence of any other ‘danger signal’. T cells reactive to such spMHCs would therefore be prone to

Table 1. Parameter values used for the calculation of the probability of escape from negative selection[a]

Parameters[b]	Description	Used values	Refs
T	Total length of the negative selection phase	14 days	[18,19]
t	Time needed to scan one DC	5 min	[20,21]
$\Rightarrow n = T/t$	Number of DCs scanned during negative selection	4000	
N_M	Total number of spMHC on the surface of one DC	5×10^5	[11]
h	Number of cognate spMHC necessary to mediate negative selection	15, 150 or 1500[c]	[11–14]
$\Rightarrow C_P = N_M/h$	Number of spMHC presented by one DC above the threshold for selection	33333, 3333 or 333	
N_P	Total number of self-peptides of 8–10 amino acids	3×10^7	SWISS-PROT/TrEMBL[d]
P_P	Probability that a given MHC allele presents the peptide	0.03	[17]
N_A	Number of MHC class I alleles in heterozygous mice	6	[18]
$\Rightarrow C_T = N_P \times P_P \times N_A$	Total number of presentable spMHC	5.4×10^6	
$\Rightarrow P = C_P/C_T$	Average probability that an spMHC is presented by a DC above the threshold for negative selection	6×10^{-3}, 6×10^{-4} or 6×10^{-5}	
$\Rightarrow P_E = (1-P)^n$	Probability of escape from negative selection	3.5×10^{-11}, 0.09 or 0.79	

[a]Abbreviations: DC, dendritic cell; spMHC, self-peptide–MHC complex.
[b]Derived expressions are indicated by arrows.
[c]The number of cognate self-peptide–MHC complexes necessary to mediate negative selection is varied as 15, 150 and 1500. The derived values, C_p, P and P_E, are provided accordingly.
[d]Downloaded from http://www.ebi.ac.uk/proteome/MOUSE/download.html. Owing to the availability of data, we have used parameter estimates from the murine immune system but the conclusions probably also apply for the human system.

Box 1. Number of selecting peptides

If a thymocyte encounters S different self-peptide–MHC complexes (spMHC), the probability of surviving selection, that is, of not reacting strongly to any of the S selecting complexes can be written as $P_S = (1 - r)^S \cong e^{-rS}$, where r is the pre-selection probability that a thymocyte is reactive to a given spMHC [31,32]. We obtain $rS \cong -\ln(P_S)$, which implies an inverse correlation between the number of selecting complexes and the reactivity required to delete a fixed fraction of lymphocytes. Approximately half to two-thirds of positively selected $CD4^+$ T cells are deleted by negative selection [33]. The probability that a naïve $CD8^+$ T cell responds to an MHC class I-restricted epitope has been estimated to be $r \approx 5 \times 10^{-6}$ [34,35]. Using this for thymocytes, and considering a deletion probability of two-thirds, yields $S = 2.2 \times 10^5$ selecting spMHC. This is consistent with the number of distinct peptides (10^4) eluted from the surface of antigen-presenting cells [36], which can be regarded as a minimum estimate owing to the limited sensitivity of detection. Using a more complicated model, Detours *et al.* have also estimated 10^3–10^5 selecting specificities [37]. It is not clear whether pre-selection reactivity of thymocytes should differ markedly from post-selection reactivity to foreign peptides [38]. A higher reactivity, and also a higher probability of surviving negative selection, would decrease the estimate for the number of selecting peptides.

Negative selection is thus restricted to ≤4% of all autoreactive specificities. T cells specific for the remaining 96% of the 5.4×10^6 spMHCs are free to leave the thymus. Interestingly, with a pre-selection reactivity of $r = 5 \times 10^{-6}$, and $C_T = 5.4 \times 10^6$ spMHC, every thymocyte is expected to be reactive to $rC_T \approx 27$ complexes on average. That is, virtually all T cells might be potentially autoreactive. This also implies that the major mechanism of peripheral tolerance cannot be irreversible anergy or apoptosis because the deletion of all autoreactive cells would mean the deletion of nearly all T cells. Instead, the reactivity of autoreactive T cells might be downregulated by 'tuning' in the peripheries [39,40].

assume effector function and cause tissue damage. In our previous calculations we have assessed the potential of thymic DCs to select against all self-peptides. However, thymic DCs seem to express a limited set of proteins related to their specific function (MHC-related proteins) and ubiquitous 'housekeeping' genes [5]. This set, which might be called the 'DC-self', is probably common to all DCs and the spMHCs produced from these proteins probably comprise a major fraction of the selecting subset. Cross-presentation of exogenous self-peptides might increase the diversity of presentation in the peripheries [4,26,27] but these changes can be regarded as additions to the basic subset of the DC-self, which is presented at all sites. Although, in principle, thymic DCs could present the sum of all peripheral DC presentation profiles, it might not be worthwhile to globally delete the T cells specific for local spMHCs. The spMHC array presented by DCs is, therefore, unlikely to go much beyond what is directly related to DC function. Other cell types in the thymus, such as mTECs, could also remove crucial T-cell specificities but their contribution to tolerance probably rests in mechanisms other than negative selection (Box 2). We therefore conclude that negative selection probably removes only those thymocytes that are reactive to spMHCs produced by DCs.

Our views are in fact at the intersection of the hypotheses of Matzinger, who first proposed that negative selection might be confined to the peptides presented by APCs [2] and Zinkernagel, who, by arguing against the relevance of 'cross-presentation' [3], identifies the peptides presented by DCs as the peptides produced by these cells – the DC-self. However, our arguments remain valid even if cross-presentation is relevant [4,26,27]. Outside the DC-self, cross-presented self-peptides would be tolerized by peripheral mechanisms, whereas those that are not cross-presented would simply be ignored [28]. Either way, they should not be involved in negative selection.

Interestingly, the theoretical work of van den Berg *et al.* [29] has shown that selection against a limited set of spMHCs can also be an efficient solution for the generation of a T-cell repertoire that can reliably distinguish between a normal self-presentation profile and the presentation profile of an infected cell.

Finally, although the framework for tolerance outlined in our Opinion is valid for both $CD8^+$ and $CD4^+$ T cells, the 'DC-self' might be less well defined for MHC class-II restricted presentation. In the case of MHC class I presentation, peptides produced by DCs are presented by both thymic and peripheral DCs, although the expression patterns might change during the maturation of peripheral DCs under inflammatory conditions [30]. By contrast, such a well-defined basic set might not exist for the MHC class II processing pathway, which acquires peptides for negative selection from the thymic environment. As a consequence, we predict that $CD4^+$ thymocytes might either survive negative selection with a higher probability than $CD8^+$ thymocytes, or could show broader cross-reactivity. Such a difference in cross-reactivity could arise, for example, if coreceptor signaling were stronger through the CD4 than through the CD8 coreceptor. The main predictions of our analysis are summarized in Box 3.

Concluding remarks

The majority of self-peptides are probably exempted from negative selection. The duration of thymic 'education' and

Box 2. Medullary thymic epithelial cells (mTECs) in central tolerance

mTECs express tissue-specific proteins in a 'promiscuous', seemingly random manner, and this mechanism has an important role in tolerance [5,6]. Although mTECs could induce negative selection in a transgenic system [41], their main contribution to tolerance cannot be deletion. Epithelial cells in the thymus express $\sim 5 \times 10^4$ MHC class I molecules on their surface, which is only 10% of the density on dendritic cells (DCs) [11]. Scanning mTECs instead of DCs would, thus, actually reduce the number of self-peptide–MHC complexes (spMHCs) scanned by a thymocyte during the available time. If DCs cannot select against all spMHCs, mTECs cannot accomplish this feat, either.

The role of mTECs in tolerance is more probably mediated by the generation of regulatory T cells [42], which has been demonstrated to be linked to thymic epithelium [43,44]. As opposed to negative selection (resulting in 'recessive tolerance'), this mechanism does not require that all thymocytes of a given self-specificity encounter their cognate spMHC. The encounter of small numbers of autoreactive thymocytes with their cognate spMHC in the random array presented by mTECs would suffice to generate regulatory cells and thus establish 'dominant tolerance'.

Box 3. Main predictions of the quantitative analysis

- The number of selecting self-peptide–MHC complexes (spMHCs) must be orders of magnitude lower than the total number of spMHCs to ensure reliable negative selection.
- Because of their lower abundance and lower MHC expression (compared to dendritic cells), medullary thymic epithelial cells cannot extend the scope of negative selection substantially. The thymic 'self-shadow' is probably involved in other mechanisms of T-cell tolerance.
- As a result of cross-reactivity, the thymocytes surviving negative selection are expected to be potentially reactive to tens of spMHCs that are not part of the selecting subset.

the limited capacity of the presentation machinery do not enable reliable selection against all potential self-specificities. Furthermore, cross-reactivity implies that all T cells might be potentially autoreactive, yet not all of them are, or should be, removed. We thus conclude that tolerance by negative selection is confined to a subset of all self-peptides. As has been argued previously by others, this subset might comprise the peptides produced by DCs.

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